

Automated Oral Disease Classification

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Abstract—Oral diseases constitute a significant global health concern and require timely and accurate diagnosis to prevent long-term complications and improve patient outcomes. Conventional diagnostic approaches rely heavily on manual inspection of oral cavity images by dental professionals, which can be subjective, time-consuming, and dependent on clinical expertise. These limitations highlight the need for reliable automated diagnostic systems to support clinical decision-making.

This paper presents a deep learning-based framework for automated multiclass classification of oral diseases using transfer learning. A comparative study is conducted using EfficientNet-B4, EfficientNet-B5, and a proposed DenseNet-169 architecture to evaluate their suitability for fine-grained oral disease recognition. To address challenges such as class imbalance and high visual similarity between disease categories, the framework incorporates class-balanced loss functions, extensive data augmentation, and a structured multi-stage fine-tuning strategy. The proposed approach aims to learn robust and discriminative feature representations from oral cavity images across six clinically relevant disease classes.

The presented framework demonstrates the potential of modern deep learning architectures to enhance automated oral disease classification and provides a foundation for developing effective computer-aided diagnostic systems for oral healthcare applications.

Index Terms—Oral Disease Classification, Deep Learning, EfficientNet, DenseNet, Medical Image Analysis, Transfer Learning

I. INTRODUCTION

Oral diseases such as dental caries, gingivitis, calculus, mouth ulcers, tooth discoloration, and hypodontia affect a large portion of the global population and require timely diagnosis to prevent serious health complications. Conventional diagnostic practices primarily rely on manual visual examination by dental professionals, which can be subjective, time-consuming, and influenced by individual clinical experience.

With recent advancements in deep learning, convolutional neural networks (CNNs) have demonstrated strong capabilities in medical image analysis by automatically learning discriminative features from visual data. Transfer learning, in particular, enables effective model training even when labeled

medical datasets are limited. Modern CNN architectures such as EfficientNet and DenseNet offer scalable and efficient feature extraction, making them well suited for analyzing high-resolution oral cavity images.

In this work, a comparative study is conducted using EfficientNet-B4, EfficientNet-B5, and a proposed DenseNet-169 model for multiclass oral disease classification. The analysis focuses on architectural characteristics, training strategies, and classification performance to identify an effective deep learning solution for automated oral disease diagnosis.

II. DATASET DESCRIPTION

The dataset used in this study comprises a total of 12,320 oral cavity images collected from multiple publicly available sources. The images represent a diverse range of oral conditions captured under varying illumination, resolution, and imaging settings, reflecting real-world clinical variability. Each image is carefully organized into one of six clinically relevant categories based on visible pathological characteristics.

The six disease classes included in the dataset are:

- Calculus
- Dental Caries
- Gingivitis
- Mouth Ulcer
- Tooth Discoloration
- Hypodontia

An inherent class imbalance is observed across the dataset, with certain disease categories being more prevalent than others. To mitigate the impact of this imbalance during training, a class-weighted focal loss function is employed, enabling the models to focus more effectively on underrepresented classes and hard-to-classify samples.

To ensure robust and unbiased evaluation, the dataset is divided using a stratified splitting strategy, preserving class distributions across all subsets. Specifically, 70% of the data is allocated for training, 15% for validation, and the remaining 15% for testing. This split allows the models to learn effectively while providing reliable performance assessment on

unseen data. The curated dataset serves as a strong foundation for evaluating the effectiveness of deep learning models in automated oral disease classification.

III. PROPOSED METHODOLOGY

A. Preprocessing and Data Augmentation

Prior to training, all input images undergo a series of pre-processing steps to ensure consistency and compatibility with the selected deep learning architectures. Each image is resized to match the input resolution required by the corresponding model, while preserving essential visual features. Pixel intensities are normalized to the range $[0, 1]$ to stabilize gradient updates and facilitate faster convergence during training.

To enhance the model's ability to generalize to unseen data and reduce overfitting, extensive data augmentation is applied exclusively to the training set. Augmentation techniques include random horizontal and vertical flipping, random brightness adjustments, contrast variation, and saturation changes. These transformations simulate realistic variations in imaging conditions such as lighting differences and viewpoint changes commonly encountered in clinical settings.

By introducing controlled variability into the training data, the augmentation strategy encourages the models to learn robust and discriminative features rather than memorizing specific visual patterns. This preprocessing and augmentation pipeline plays a crucial role in improving classification performance, particularly for underrepresented classes and visually similar oral disease categories.

B. Network Architecture

The proposed architecture follows a modular transfer learning design that combines a powerful pretrained feature extractor with a lightweight task-specific classification head. Each input oral cavity image is first resized and normalized to ensure consistency across models, followed by on-the-fly data augmentation to improve robustness to variations in lighting, orientation, and imaging conditions.

A pretrained convolutional backbone—EfficientNet-B4, EfficientNet-B5, or DenseNet-169—is then used to extract high-level semantic features from the input image. These backbones are initialized with ImageNet weights, enabling effective feature reuse and faster convergence. The extracted feature maps are passed through a global average pooling layer, which reduces spatial dimensions while preserving discriminative information and minimizing overfitting.

Subsequently, batch normalization is applied to stabilize training, followed by a fully connected layer with 512 neurons and ReLU activation to learn task-specific representations. A dropout layer with a rate between 0.4 and 0.5 is used to improve generalization by reducing co-adaptation among neurons. Finally, a softmax output layer produces class probabilities corresponding to the six oral disease categories. This unified architectural design enables fair comparison across models while allowing each backbone to leverage its intrinsic feature learning strengths.

Figure 2 illustrates the overall architecture adopted across all models.

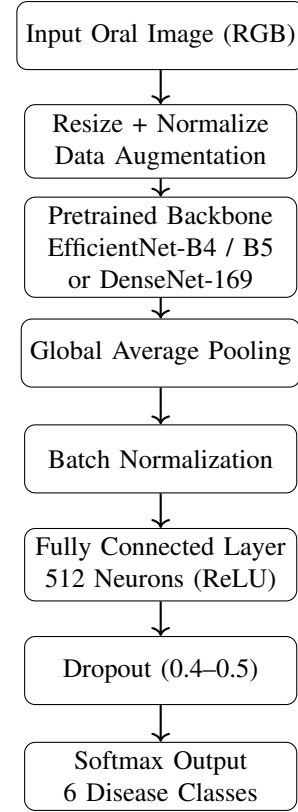


Fig. 2: Overall architecture of the proposed oral disease classification framework

C. Loss Function and Optimization

The oral disease dataset used in this study exhibits notable class imbalance, with conditions such as calculus and hypodontia appearing less frequently than others like dental caries and mouth ulcers. Training deep neural networks with standard cross-entropy loss under such imbalance often results in biased predictions toward majority classes. To address this challenge, a class-aware loss strategy is employed.

During the initial training phase, a class-weighted sparse categorical cross-entropy loss is used. Class weights are computed based on inverse class frequencies and applied directly to the loss function, ensuring that underrepresented disease categories receive higher importance during optimization. This approach stabilizes early training, particularly when the pre-trained backbone remains frozen.

In later stages, the loss function is enhanced by incorporating focal loss, which emphasizes hard-to-classify and visually ambiguous samples by down-weighting well-classified examples. This is especially beneficial for oral disease images, where inter-class similarities are common. The combination of class weighting and focal loss enables effective handling of both class imbalance and sample difficulty.

Model optimization is carried out using the Adam optimizer due to its adaptive learning rate and robust convergence be-



Fig. 1: samples for the six oral disease classes used in this study: caries, gingivitis, tooth discoloration, ulcers, hypodontia, and calculus.

havior. A staged learning rate schedule is applied, with higher learning rates during classifier training and lower rates during backbone fine-tuning. The ReduceLROnPlateau scheduler and early stopping based on validation loss are employed to ensure stable convergence and prevent overfitting. Gradual unfreezing of pretrained layers further facilitates effective domain adaptation while preserving essential low-level visual features.

D. Fine-Tuning Strategy

To effectively adapt pretrained convolutional networks to the oral disease classification task, a multi-stage fine-tuning strategy is employed. This approach balances the preservation of robust generic visual features learned from large-scale datasets with the need to capture domain-specific characteristics present in oral cavity images.

- 1) In the first stage, the pretrained backbone network is fully frozen, and only the newly added classification head is trained. This allows the dense layers to learn high-level discriminative representations specific to oral disease categories without disturbing the pretrained feature extractor. During this phase, a relatively higher learning rate is used to accelerate convergence.
- 2) In the second stage, gradual unfreezing is applied, where only the upper layers of the backbone are made trainable while deeper layers remain frozen. This controlled unfreezing enables the model to refine mid- and high-level features that are more task-specific, such as texture variations and lesion boundaries, while retaining stable low-level representations like edges and color gradients.
- 3) In the final stage, a low learning rate fine-tuning process is conducted by unfreezing a larger portion of the backbone network. A reduced learning rate is crucial at this stage to prevent catastrophic forgetting and ensure smooth adaptation of pretrained weights. This progressive training strategy significantly improves generalization performance and contributes to stable optimization, particularly for visually similar oral disease classes.

IV. EXPERIMENTAL RESULTS

A. Classification Performance

Table I summarizes the performance of all models on the test set.

TABLE I: Performance Comparison of Models

Model	Accuracy	Macro F1	Weighted F1
EfficientNet-B4	86.42%	85.59%	86.61%
EfficientNet-B5	90.37%	89.14%	90.44%
DenseNet-169 (Proposed)	95.29%	91.91%	93.28%

The results in Table I clearly indicate notable performance differences among the evaluated models. EfficientNet-B4 serves as a strong baseline, achieving reasonable accuracy and F1-scores across all classes; however, its performance is limited when handling visually similar oral disease categories. EfficientNet-B5 improves upon this baseline by leveraging deeper and wider feature representations, resulting in a significant gain in both accuracy and F1 metrics.

The proposed DenseNet-169 model demonstrates the best overall performance across all evaluation criteria. The substantial improvement in macro and weighted F1-scores indicates more balanced and reliable classification across all disease classes, including underrepresented and challenging categories. This performance gain can be attributed to DenseNet's dense connectivity, which facilitates effective feature reuse and enhances gradient flow during training.

Overall, the quantitative results confirm that deeper architectures with dense feature propagation are better suited for fine-grained oral disease classification, where subtle texture and structural variations play a crucial role.

B. Classification Reports

Figure 3, Figure 4, and Figure 5 present the classification reports for EfficientNet-B4, EfficientNet-B5, and the proposed DenseNet-169 model, respectively. These reports provide detailed insights into class-wise precision, recall, and F1-score for each oral disease category.

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===== EfficientNet-B4 Classification Report =====

              precision    recall  f1-score   support

    Calculus      0.579      0.774      0.662       195
     Caries      0.898      0.923      0.910       390
    Gingivitis    0.832      0.688      0.753       352
    Mouth_Ulcer    0.949      0.933      0.941       421
Tooth_Discoloration 0.919      0.904      0.912       303
     Hypodontia    0.967      0.947      0.957       187

   accuracy              0.864       1848
  macro avg      0.857      0.862      0.856       1848
 weighted avg      0.874      0.864      0.866       1848

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Fig. 3: Classification report for EfficientNet-B4

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===== EfficientNet-B5 Classification Report =====

              precision    recall  f1-score   support

    Calculus      0.668      0.703      0.685       195
     Caries      0.944      0.946      0.945       390
    Gingivitis    0.817      0.812      0.815       352
    Mouth_Ulcer    0.986      0.976      0.981       421
Tooth_Discoloration 0.950      0.944      0.947       303
     Hypodontia    0.984      0.968      0.976       187

   accuracy              0.904       1848
  macro avg      0.891      0.892      0.891       1848
 weighted avg      0.905      0.904      0.904       1848

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Fig. 4: Classification report for EfficientNet-B5

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===== DenseNet-169 Classification Report =====

              precision    recall  f1-score   support

    Calculus      0.720      0.713      0.716       195
     Caries      0.977      0.995      0.986       390
    Gingivitis    0.836      0.841      0.839       352
    Mouth_Ulcer    0.995      0.998      0.996       421
Tooth_Discoloration 1.000      0.987      0.993       303
     Hypodontia    0.995      0.973      0.984       187

   accuracy              0.933       1848
  macro avg      0.921      0.918      0.919       1848
 weighted avg      0.933      0.933      0.933       1848

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Fig. 5: Classification report for the proposed DenseNet-169 model

C. Confusion Matrix Analysis

The DenseNet-169 model demonstrates superior class-wise discrimination, particularly for visually similar categories such as calculus and gingivitis. Misclassification rates are significantly reduced compared to EfficientNet variants.

Confusion matrices provide a detailed view of class-wise prediction behavior and misclassification patterns. Figures 6, 7, and 8 illustrate the confusion matrices obtained for EfficientNet-B4, EfficientNet-B5, and DenseNet-169 models, respectively.

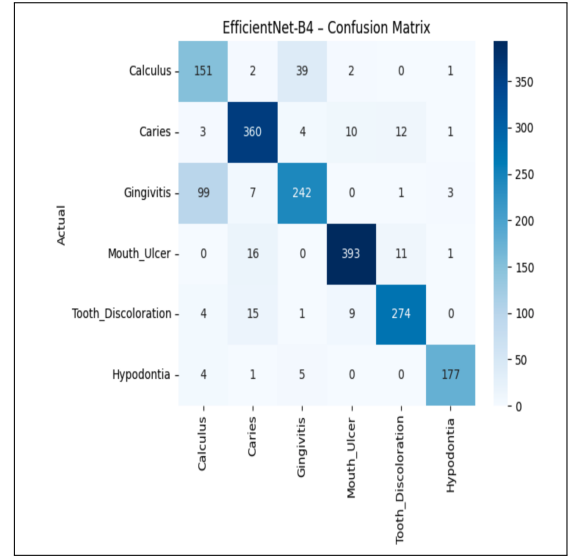


Fig. 6: Confusion matrix for EfficientNet-B4

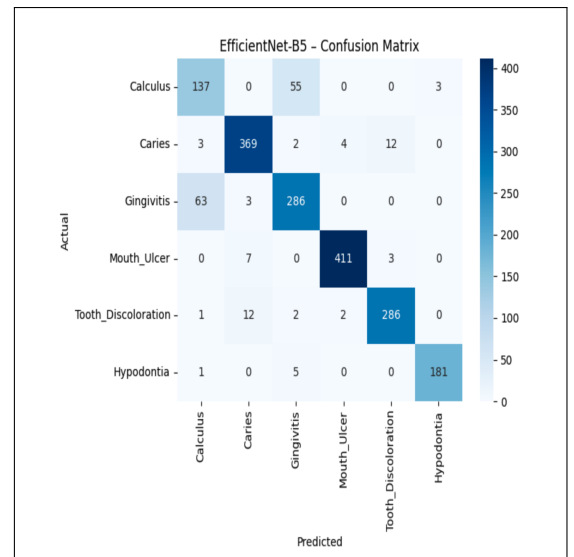


Fig. 7: Confusion matrix for EfficientNet-B5

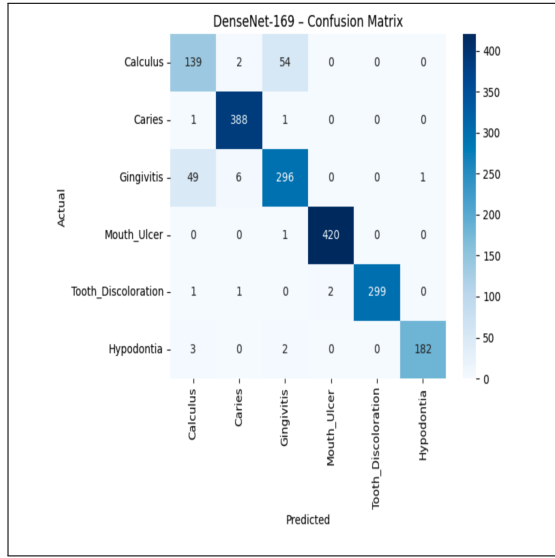


Fig. 8: Confusion matrix for the proposed DenseNet-169

The DenseNet-169 model demonstrates reduced confusion among visually similar disease classes such as calculus and gingivitis, highlighting its superior discriminative capability.

D. Discussion

DenseNet-169 benefits from its dense connectivity, where each layer receives feature maps from all preceding layers. This design improves gradient flow and promotes effective feature reuse, enabling the network to learn stable and discriminative representations while mitigating vanishing gradient issues common in deep architectures.

For oral disease classification, this property is particularly beneficial due to the subtle visual differences among disease categories such as calculus and gingivitis. DenseNet-169 effectively combines low-level texture information with high-level semantic features, leading to improved class separability. Additionally, the dense connections facilitate efficient transfer learning by allowing pretrained features to be reused and adapted without significant information loss. As a result, the model demonstrates reduced misclassification and more balanced performance across all disease classes.

E. Model Comparison Analysis

To visually compare the overall performance of the evaluated models, comparison plots are generated using accuracy and weighted F1-score metrics.

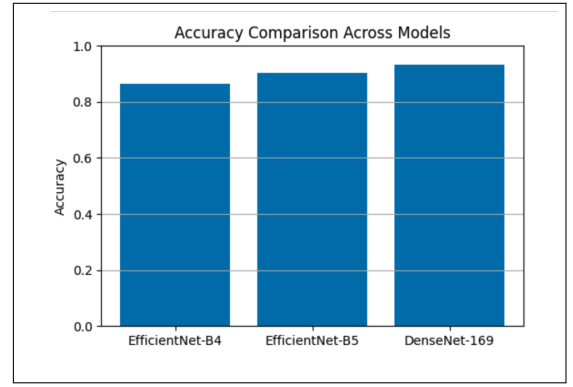


Fig. 9: Accuracy comparison across EfficientNet-B4, EfficientNet-B5, and DenseNet-169

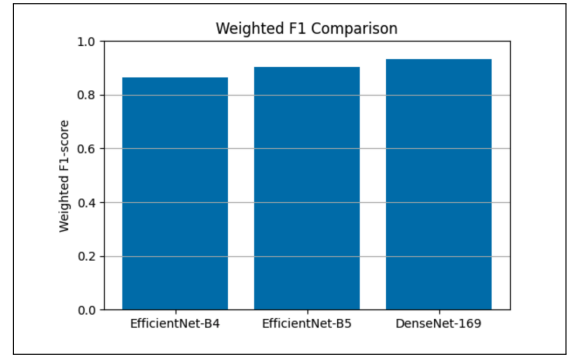


Fig. 10: Weighted F1-score comparison across models

These plots clearly indicate that the proposed DenseNet-169 model consistently outperforms EfficientNet variants across all evaluation metrics.

V. CONCLUSION

This paper presented a comprehensive deep learning-based framework for automated multiclass oral disease classification using transfer learning. A systematic comparative study was conducted using EfficientNet-B4, EfficientNet-B5, and a proposed DenseNet-169 architecture to evaluate their effectiveness in distinguishing six clinically significant oral disease categories.

Experimental results demonstrate that the proposed DenseNet-169 model consistently outperforms the EfficientNet-based models across all evaluation metrics, achieving the highest accuracy, macro F1-score, and weighted F1-score. The superior performance of DenseNet-169 can be attributed to its dense connectivity pattern, which promotes effective feature reuse, improved gradient propagation, and enhanced learning of subtle texture and structural variations present in oral cavity images. These characteristics are particularly important for differentiating visually similar disease classes such as calculus and gingivitis.

The incorporation of class-balanced loss functions, focal loss, extensive data augmentation, and a multi-stage fine-tuning strategy further contributed to robust model general-

ization and stable optimization under class-imbalanced conditions. The results indicate that deep feature reuse combined with controlled fine-tuning enables reliable performance even in challenging medical imaging scenarios.

Overall, the findings highlight the potential of the proposed DenseNet-169-based framework as an effective computer-aided diagnostic tool for oral healthcare applications. Future work will focus on integrating explainable artificial intelligence techniques to enhance clinical interpretability, validating the system on larger multi-center datasets, and exploring real-time deployment in clinical decision support systems.

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